



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF SCIENCE**

Postgraduate Programme
(Courses effective from Academic Year 2018-19)

SYLLABUS OF COURSES TO BE OFFERED
“Master of Science”
(Mathematics)

Disclaimer: The postgraduate programme in Master of Science Syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Science, Bhabha University

Syllabus and Scheme

M.Sc Mathematics

ALL SEMESTERS

S. No.	Subject Code	Subject Name	Marks Allotted							
			Assignment Marks		Theory		Practical		Total Marks	
			Max	Min	Max	Min	Max	Min	Max	Min
SEMESTER-I										
1	MMT-101	Advanced Abstract Algebra-I	80	29	20	7	0	0	100	36
2	MMT-102	Real Analysis	80	29	20	7	0	0	100	36
3	MMT-103	Topology-I	80	29	20	7	0	0	100	36
4	MMT-104	Complex Analysis-I	80	29	20	7	0	0	100	36
5	MMT-105	Advanced Discrete Mathematics-I	80	29	20	7	0	0	100	36
SEMESTER-II										
1.	MMT-201	Advanced Abstract Algebra –II	80	29	20	7	0	0	100	36
2.	MMT-202	Lebesgue Measure and Integration	80	29	20	7	0	0	100	36
3.	MMT-203	Topology-II	80	29	20	7	0	0	100	36
4.	MMT-204	Complex Analysis-II	80	29	20	7	0	0	100	36
5.	MMT-205	Advanced Discrete Mathematics-I	80	29	20	7	0	0	100	36
SEMESTER-III										
1.	MMT-301	Functional Analysis-I	80	29	20	7	0	0	100	36
2.	MMT-302	Advanced Special Function-I	80	29	20	7	0	0	100	36
3.	MMT-303	Numerical Analysis	80	29	20	7	0	0	100	36
4.	MMT-304	Spherical Trigonometry and Astronomy	80	29	20	7	0	0	100	36
5.	MMT-305	Theory of Linear Operators - I	80	29	20	7	0	0	100	36
SEMESTER-IV										
1.	MMT-401	Functional Analysis-II	80	29	20	7	0	0	100	36
2.	MMT-402	Integral Transform	80	29	20	7	0	0	100	36
3.	MMT-403	Advanced Graph Theory	80	29	20	7	0	0	100	36
4.	MMT-404	Operation Research	80	29	20	7	0	0	100	36
5.	MMT-405	ELECTIVE								

ELECTIVE [Optional Paper (choose any one)]										
1	MMT-405[I]	Theory of Linear Operators - II	80	29	20	7	0	0	100	36
2	MMT-405[II]	Fundamental of Computer(Theory & Practical)	60	22	20	7	20	7	100	36
3	MMT-405[III]	Advanced special function	80	29	20	7	0	0	100	36
4	MMT-405[IV]	Fuzzy sets and their applications	80	29	20	7	0	0	100	36
5	MMT-405[V]	Analytic number theory	80	29	20	7	0	0	100	36
6	MMT-405[VI]	Partial Differential equation	80	29	20	7	0	0	100	36
7	MMT-405[VII]	Integration theory	80	29	20	7	0	0	100	36

* w.e.f. 2019-2020

Class	-	M.Sc
Semester	-	I
Subject	-	Mathematics
Paper Title	-	Advanced Abstract Algebra–I
Paper	-	I

Max. Marks : 80

Unit-1 Normal & Subnormal series of groups, Composition series, Jordan Holder series.

Unit-2 Solvable & Nilpotent groups.

Unit-3 Extension fields. Roots of polynomials, Algebraic and transcendental extensions. Splitting fields. Separable and inseparable extension.

Unit-4 Perfect fields, Finite fields, Algebraically closed fields.

Unit-5 Automorphism of extension, Galois extension. Fundamental theorem of Galois theory Solution of polynomial equations by radicals, Insolubility of general equation of degree 7 by radicals.

Text books :-

- (1) I.N. Herstein, Topics in Algebra, ,Wiley Eastern, New Delhi.
- (2) V.Sahai & V. Bisht, Algebra, Narosa Publishing House.
- (3) P.B. Bhattacharya, S.K. Jain and S.R. Nagpaul, Basic Abstract Algebra, Cambridge University press.

Reference Books :-

- (1) N. Jacobson, Basic Algebra, Vol. I, II & VIII, Hindustan Publishing Company.
 - (2) S. Lang, Algebra, Addison- Wesley.
 - (3) I.S. Luther & I.B.S. Passi Algebra vol- 1,2,3 Narosa company.
- Books prescribed in unified syllabus.

Class	-	M.Sc
Semester	-	I
Subject	-	Mathematics
Paper Title	-	Real Analysis
Paper	-	II

Max. Marks : 80

- Unit-I Definition and existence of Riemann- Stieltjes integral and its properties, Integration and differentiation.
- Unit-II Integration of vector- valued functions, Rectifiable curves. Rearrangements of terms of a series. Riemann" s theorem.
- Unit-III Sequences and series of functions, Point wise and uniform convergence, Cauchy criterion for uniform convergence, Weierstrass M-test, uniform convergence and continuity, uniform convergence and Riemann-Stieltjes integration, uniform convergence and differentiation.
- Unit-IV Functions of several variables, linear transformations, Derivatives in an open subset of R^n , Chain rule, Partial derivatives, Differentiation, Inverse function theorem.
- Unit-V Derivatives of higher order, Power series, uniqueness theorem for power series, Abel's and Tauber's theorems, Implicit function theorem,

Text books :

1. Walter Rudin, Principles of Mathematical Analysis, McGraw Hill.

Reference books:

1. T.M. Apostol, Mathematical Analysis Narosa.
2. H.L. Royden, Real Analysis, Macmillan (Indian Edition)
3. Books prescribed in unified syllabus.

Class	-	M.Sc
Semester	-	I
Subject	-	Mathematics
Paper Title	-	Topology-I
Paper	-	III

Max. Marks : 80

Unit – I Countable and uncountable sets. Infinite sets and the Axiom of Choice. Cardinal numbers and its arithmetic. Schroeder-Bernstein theorem, statements of Cantor's theorem and the Continuum hypothesis. Zorn's lemma. well- ordering theorem. [G.F. Simmons and K.D. Joshi]

Unit- II Definition and examples of topological spaces. Closed sets. Closure. Dense subsets. Neighbourhoods, interior exterior and boundary. Accumulation points *and* derived sets. Bases and sub-bases, Subspaces and relative topology. [G.F. Simmons]

Unit-III Alternate methods of defining a topology in terms of Kuratowski Closure Operator and Neighbourhood Systems. Continuous functions and homeomorphism. [G.F. Simmons, K.D. Joshi, J.R. Munkers]

Unit- IV First and Second Countable spaces. Lindelof' s theorems. Separable spaces. Second Countability and Separability. [G.F., Simmons]

Unit- V Path-connectedness, connected spaces. Connectedness on Real line. Components, Locally connected spaces. [J.R. Munkers]

Text Books:-

1. J.R. Munkers, Topology-A first course, Prentice-Hall of India.
2. G.F. Simmons, Introduction to Topology and Modern Analysis, McGraw Hill
3. K.D. Joshi: Introduction to General Topology, Wiley Eastern.

Books prescribed in unified syllabus.

Class	-	M Sc
Semester	-	I
Subject	-	Mathematics
Title	-	Complex Analysis-I
Paper	-	IV

Max. Marks : 80

Unit-I Complex integration, Cauchy – Goursat theorem, Cauchy integral formula, Higher order derivatives

Unit-II Morera" s theorem. Cauchy" s inequality. Liouville" s theorem. The fundamental theorem of algebra. Taylor" s theorem.

Unit-III The maximum modulus principle. Schwartz lemma. Laurent series. Isolated singularities. Meromorphic functions, The argument principle. Rouché" s theorem. Inverse function theorem.

Unit – IV Residues. Cauchy" s residue theorem. Evaluation of integrals. Branches of many valued functions with special reference to $\arg z, \log z, z^a$.

Unit – V Bilinear transformations, their properties and classification. Definitions and examples of conformal mappings.

Text Book :

1. J.B. Conway, Functions of one complex variable, Springer-verlag

Reference Books:

1. S. Ponnuswamy, Foundations of complex analysis, Narosa Publishing House.
2. L.V. Ahlfors, Complex analysis, McGraw Hill
3. Books prescribed in unified syllabus.

Class	-	M Sc
Semester	-	I
Subject	-	Mathematics
Paper Title	-	Advanced Discrete Mathematics-I
Paper	-	V

Max. Marks : 80

Unit-I Semigroups and monoids, subsemigroups and submonoids, Homomorphism of semigroups and monoids, Congruence relation and Quotient semigroups, Direct products, Basic Homomorphism Theorem.

Unit-II Lattices- Lattices as partially ordered sets, their properties, Lattices as Algebraic systems, sublattices, Bounded lattices, Distributive Lattices, Complemented lattices.

Unit-III Boolean Algebra- Boolean Algebras as lattices, various Boolean identities. Joint irreducible elements, minterms, maxterms, minterm Boolean forms, canonical forms, minimization of Boolean functions. Applications of Boolean Algebra to switching theory (Using AND, OR, & NOT gates) the Karnaugh method.

Unit-IV Graph Theory- Definition and types of graphs. Paths & circuits. Connected graphs. Euler graphs, weighted graphs (undirected) Dijkstra" s Algorithm. Trees, Properties of trees, Rooted & Binary trees, spanning trees, minimal spanning tree.

Unit-V Complete Bipartite graphs, Cut-sets, properties of cut sets, Fundamental Cut-sets & circuits, Connectivity and Separability, Planar graphs, Kuratowski" s two graphs, Euler" s formula for planar graphs.

Text Books :-

1. J.P. Tremblay & R. Manohar, Discrete Mathematical structures, McGraw Hill.
2. N. Deo. : Graph theory with applications Prentice hall.

Reference Books:-

1. C.L. Liu : Elements of Discrete Mathematics McGraw Hill.
2. Semyour Lipschutz/More lipson : Discrete Mathematics, McGraw Hill.
3. Books prescribed in unified syllabus

Class	-	M.Sc	
Semester	-	II	
Subject	-	Mathematics	
Paper Title	-	Advanced Abstract Algebra-II	
Paper	-	I	Max. Marks : 80

Unit-1 Introduction to Modules, examples, submodules, quotient modules, Module homomorphism, isomorphism, Finitely generated modules, cyclic modules.

Unit-2 Simple modules, Semisimple modules, Free modules, Schur's lemma.

Unit-3 Noetherian & Artinian modules and rings, Hilbert basis theorem. Wedderburn -Artin theorem.

Unit-4 Uniform modules, Primary modules, Noether-Laskar theorem, Fundamental structure theorem of modules over a principal ideal domain.

Unit-7 Algebra of Linear Transformation, characteristic roots, Matrices, Matrix of Linear Transformations, similarity of Linear transformation, Invariant spaces, Reduction to triangular forms.

Text book :

- (1) P.B. Bhattacharya, S.K. Jain, S.R. Nagpaul, Basic Abstract Algebra, Cambridge University Press, (Indian Edition)
- (2) I.N. Herstein, Topics in Algebra, Wiley Eastern.
- (3) S. Kumaresan, Linear Algebra-A geometric approach, Prentice Hall India Ltd.
- (4) Books prescribed in unified syllabus.

Class	-	M Sc
Semester	-	II
Subject	-	Mathematics
Paper Title	-	Lebesgue Measure & Integration
Paper	-	II

Max. Marks : 80

Unit –I Lebesgue outer measure. Measurable sets. Regularity. Measurable functions. Borel and Lebesgue measurability. Non-measurable sets.

Unit – II Integration of Non-negative functions. The General integral. Integration of Series, Riemann and Lebesgue Integrals.

Unit – III The Four derivatives. Functions of Bounded variation. Lebesgue Differentiation Theorem, Differentiation and Integration.

Unit – IV The L^p -spaces, Convex functions, Jensen's inequality. Holder and Minkowski inequalities. Completeness of L^p .

Unit – V Dual of space when $1 < p < \infty$, convergence in measure, Uniform. Convergence and almost uniform convergence.

Text book :

1. G. de Barra. Measure Theory and Integration, Wiley Eastern (Indian Edition)

References Books:

1. Walter Rudin, Principles of Mathematical Analysis, McGraw-Hill, International student edition,
2. H.L. Royden, Real Analysis, Macmillan, Indian Edition.
3. Books prescribed in unified syllabus.

Class	-	M Sc
Semester	-	II
Subject	-	Mathematics
Paper Title	-	Topology II
Paper	-	III

Max. Marks : 80

Unit – I Separation Axioms T_0 , T_1 , T_2 , T_3 , T_4 Their Characterizations and basic properties, Urysohn's lemma. Tietze Extension Theorem. [K.D. Joshi, G.F. Simmons]

Unit – II Compactness. Continuous functions and compact sets. Basic properties of compactness. Compactness and finite intersection property. Sequentially and countably compact sets, Local compactness. [G.F. Simmons]

Unit –III Tychonoff product topology in terms of standard sub-base and its characterizations. Projection maps. Separation axioms and product spaces, Connectedness and product spaces, Compactness and product spaces (Tychonoff's theorem) Countability and product spaces. [G.F. Simmons]

Unit – IV Net and filter's Topology and convergence of nets, Hausdorffness and nets, Compactness and nets. Filters and their convergence, Canonical way of converting nets to filters and vice-versa. Ultrafilters and Compactness. [K.D. Joshi]

Unit –V The fundamental group and covering spaces-Homotopy of paths, The fundamental group, Covering spaces, The fundamental group of the circle and the fundamental theorem of Algebra. [J.R. Munkers]

Text books :-

1. James R. Munkres, Topology, A First Course, Prentice Hall of India Pvt. Ltd. New Delhi.
2. G.F. Simmons, Introduction to Topology and Modern Analysis, McGraw-Hill Book Company.
3. K.D. Joshi, Introduction to general topology Wiley Eastern Ltd.
4. Books prescribed in unified syllabus.

Class	-	M.Sc
Semester	-	II
Subject	-	Mathematics
Paper Title	-	Complex Analysis II
Paper	-	IV

Max. Marks : 80

Unit – I Weierstrass factorization theorem. Gamma and its properties. Riemann Zeta function. Riemann's functional equation

Unit – II Mittag-Leffler's theorem. Analytic continuation. Uniqueness of direct analytic continuation. Uniqueness of analytic continuation along a curve. Power series method of analytic continuation.

Unit – III Schwartz reflection principle. Harmonic function on a disc. Harnack inequality and theorem. Dirichlet problem. Green's function.

Unit-IV Canonical products. Jensen's formula. Hadamard's three circles theorem. Order of an entire function. Exponent of convergence. Borel's theorem Hadamard's factorization theorem.

Unit-V The range of an analytic function. Bloch's theorem. The little Picard theorem. Schottky's theorem. Montel Caratheodory and great Picard theorem. Univalent function. Bieberbach conjecture and the $\frac{1}{4}$ - theorem.

Text Book.:-

1. J.B. Conway, Functions of one complex variable, Springer-Verlag

Reference

1. S Ponnuswamy, Fundamentals of complex analysis, Narosa Publishing House.
2. L.V. Ahlfors, Complex Analysis, McGraw Hill

Class	-	M.Sc
Semester	-	II
Subject	-	Mathematics
Paper Title	-	Advanced Discrete Mathematics II
Paper	-	V

Max. Marks : 80

Unit-I Matrix representation of graphs- Incidence matrix, Cut set matrix, Path Matrix, Circuit Matrix and Adjacency matrix. Directed graphs Definition of types of directed graphs, Binary search trees.

Unit-II Discrete Numeric functions, Asymptotic Behaviour of Numeric functions, Generating functions. Recurrence Relations- Linear Recurrence Relations with constant coefficients, Homogeneous solutions, particulars solutions, Total Solutions.

Unit-III Computability and Formal languages- Languages, Phrase structure grammars, Derivation, Sentential forms, Language generated by grammar. Regular, context-free and context sensitive grammars.

Unit-IV Finite State Automata, Diagram & Language determined by an Automaton, Finite State Acceptors, Deterministic and Non-deterministic Finite Automata. Finite State Machines, their transition tables & diagrams. Equivalent machines.

Unit-V Reduced Machines, Kleen's theorem (Statement only) Pumping lemma, Moore and Mealy Machines, Turning Machines. Regular Expressions and corresponding regular languages (Def only)

Text Books :-

1. J.P. Tremblay & R. Manohar; Discrete Mathematical Structures, McGraw Hill.
2. N. Deo. Graph theory with applications, Prentice Hall.

References :-

1. C.L. Liu : Elements of Discrete Mathematics McGraw Hill.
2. Semyour Lipschutz/Marc Lipson Discrete Mathematics, McGraw Hill.

Class : M.Sc.
Semester : III
Subject : Mathematics
Paper : I
Paper Title : Functional Analysis-I

Max. Marks : 80

Unit-I Normed linear spaces. Banach Spaces and examples. Properties of normed linear spaces, Basic properties of finite dimensional normed linear spaces.

Unit-II Normed linear subspace, equivalent norms, Riesz lemma and compactness.
Quotient space of normed linear spaces and its completeness.

Unit-III Linear operator, Bounded linear operator and continuous operators.

Unit-IV Linear functional, bounded linear functional, Dual spaces with examples.

Unit-V Hilbert space, orthogonal complements, orthonormal sets and sequences.
Representation of functional on Hilbert spaces.

Text Books :

1. E. Kreyszig, Introductory functional analysis with application, Jhon Wiley & sons. New York 1978.
2. G.F. Simmons, Introduction to Topology & Modern Analysis, Tata Mc Graw Hill, New York.

Reference :-

1. B. Choudhary and Sudarshan Nanda, Functional Analysis with applications, Wiley Eastern Ltd.

Class : M.Sc.
Semester : III
Subject : Mathematics
Paper : II
Paper Title : Advanced Special Function-I

Max. Marks : 80

Unit-I Gamma and Beta Function: The Euler or Macheroni Constant γ , Gamma Function
A series for $\Gamma'(z)/\Gamma(z)$, Difference equation $\Gamma(z+1) = z\Gamma(z)$

$z\Gamma(z)$, value of $\Gamma(1-z)$, Factorial function, Legendre's duplication formula,
Gauss multiplication theorem.

Unit-II Hypergeometric function and function ${}_2F_1(a,b;c;z)$. A simple integral form
valuation of ${}_2F_1(a,b;c;z)$. Contiguous function relations, Hyper geometric
differential equation and its solutions, $F(a,b;c;z)$ as function of its
parameters.

Unit-III Generalized Hypergeometric function.

Unit-IV Elementary series manipulations, Simple transformation,
Relations between functions of z and $1-z$.

Unit-V Confluent hyper geometric function and its properties.

Books Recommended :-

1. Rainville, E.D., Special Functions, the Macmillan Co., New York 1971.
2. Srivastava, H.M., Gupta K.C. and Goyal, S.P. ; The H-Functions of one and two variables with applications, South Asian Publication, New Delhi.
3. Saran N., Sharma S.D. and Trivedi- Special Functions with application, Pragati Prakashan 1986.
4. The saxena V.P.- I-Function, anamaya- New Delhi, 2008.

Reference Books :-

1. Lebedev, N.N., Special Functions and Their Applications, Prentice Hall, Englewood Cliffs, New Jersey, USA 1997.
2. Whittaker, E.T. and Watson, G.N., A course of Modern Analysis Cambridge University Press, London, 1963.

Class	:	M.Sc
Semester	:	III
Subject	:	Mathematics
Paper Title	:	Numerical Analysis
Paper	:	III

Max. Marks : 80

Unit-I Interpolation

Existence, Uniqueness of interpolating polynomial, Error of interpolation, Unequally spaced data; Lagrange's, Newton's divided difference formulae. Equally spaced data: finite difference operators and their properties, Gauss's forward and backward formulae, Inverse interpolation, Hermite interpolation.

Unit-II Differentiation

Finite difference approximations for first and second order derivatives. Integration: Newton-cotes closed type methods; Particular cases, Error terms, Newton cotes open type methods, Romberg integration, Gaussian quadrature; Legendre, Chebyshev formulae.

Unit-III Solution of nonlinear and transcendental equations

Regula-Falsi, Newton-Rapson method, Chebyshev's, method, Muller's method, Birge-Vita method, Solution of system of nonlinear equations.

Approximation : Norms, Least square (using monomials and orthogonal polynomials), Uniform and Chebyshev approximations.

Unit-IV Solution of linear algebraic system of equations

LU Decomposition, Gauss-Seidal methods; solution of tridiagonal system. Ill conditioned equations. Eigen values and Eigen vectors: Power and Jacobi methods.

Unit-V Solution of Ordinary differential equations

Initial value problems: Single step methods; Taylor's, Euler's, Runge-Kutta methods, Error analysis; Multi-step methods: Adam-Bashforth, Nystorm's, Adams- Moulton methods, Milne's predictor-corrector methods. System of IVP's and higher order IVP's.

Text Books:

1. M.K. Jain, S.R.K. Iyengar and R.K. Jain : Numerical Methods for Engineers and Scientists, New Age International, 2003.
2. C.F. Gerald, P.O. Wheatley: Applied Numerical Analysis, Addison-Wesley, 1994.

Reference Books:

1. D. Aitkinson : Numerical Analysis, John Wiley and Sons, 2009.
2. Samuel D. Cante and Carl de Boor: Elementary Numerical Analysis, McGraw Hill, 1980.

Class	-	M.Sc
Semester	-	III
Subject	-	Mathematics
Paper Title	-	Spherical Trigonometry and Astronomy
Paper	-	IV

Max. Marks : 80

- Unit-I:** Spherical Trigonometry - up to solution of right angled Triangles.
- Unit-II:** General ideas and relation between sides and Angles of a spherical triangle.
- Unit-III:** Spherical Astronomy - Various System of references and related topics.
- Unit-IV:** Celestial sphere, Transit instrument.
- Unit-V:** Atmospheric Refraction. Time planetary phenomena.

TEXTBOOKS:

1. A text book of spherical trigonometry : Gorakh Prasad.
2. A text book of spherical Astronomy : Gorakh Prasad.

REFERENCEBOOKS:

1. Spherical Astronomy - Smart.
2. Spherical Astronomy - Bell.

Class : M.Sc.
Semester : III
Subject : Mathematics
Paper Title : Theory of Linear Operators-I
Paper : V

Max. Marks : 80

Unit-I Spectral Theory in finite dimensional normed spaces.
Regular value resolvent set and spectrum.

Unit-II Spectral Properties of Bounded Linear Operators, resolvent
and spectral mapping theorem for polynomials.

Unit-III Spectral radius of a bounded linear operator on a complex Banach space.
Banach Algebra, Further properties of Banach Algebras.

Unit-IV Compact linear operators on normed spaces,
further properties of compact linear operators.

Unit-V Spectral properties of compact linear operators.

Text Books :

1. E. Kreyszig, Introductory functional analysis with application, John Wiley & sons. New York 1978.
2. G.F. Simmons, Introduction to Topology & Modern Analysis, Tata McGraw Hill, New York.

Reference :-

1. P.R. Halmos, Introduction to Hilbert space and the theory of spectral Multiplicity, second Edition, Chelsea Publishing Co. N.Y. 1977.
2. N. Dunford and J.T. Schwartz, Linear operator-3 part inter science/Wiley New York, 1978-71.
3. G. Bachman and L. Narcil, Functional analysis from academic press

Books Recommended :-

1. Rainville, E.D., Special Functions, the Macmillan Co., New York 1971.
2. Srivastava, H.M., Gupta K.C. and Goyal, S.P. ; The H- Functions of one and two variables with applications, South Asian Publication, New Delhi.
3. Saran N., Sharma S.D. and Trivedi- Special Functions with application, Pragati Prakashan 1986.
4. The saxena V.P.- I-Function, anamaya- New Delhi, 2008.

Reference Books :-

1. Lebdev, N.N., Special Functions and Their Applications, Prentice Hall, Englewood Cliffs, New Jersey, USA 1997.
2. Whittaker, E.T. and Watson, G.N., A course of Modern Analysis Cambridge University Press, London, 1963.

Class : M.Sc.
Semester : IV
Subject : Mathematics
Paper Title : Functional Analysis-II
Paper : I

Max. Marks : 80

Unit- I Hilbert adjoint operator and its properties, self adjoint, Unitary and normal operators positive operator.

Unit-II Zorn's Lemma Hahn-Banach Theorem for real linear spaces, Hahn-Banach theorem for complex linear space and normed linear spaces.

Unit-III Adjoint operators on normed spaces, relation between adjoint operator and Hilbert adjoint operator, Reflexive spaces, Reflexivity of Hilbert space.

Unit-IV Category theorem- Baire's Category theorem, uniform boundedness theorem and some of its application, strong and weak convergence in normed spaces.

Unit-V Convergence of sequences of operators and functionals, open mapping theorem, closed graph theorem, contraction theorem.

Text Books:

1. E. Kreyszig, Introductory Functional Analysis with applications, John Wiley & Sons, New York 1978.
2. G.F. Simmons, Introduction to Topology & Modern Analysis McGraw Hill, New York.

Reference :

1. B. Choudhary and Sudarshan Nanda, Functional Analysis with applications, Wiley Eastern Ltd.

Class : M.Sc.
Semester : IV
Subject : Mathematics
Paper Title : Integral Transform
Paper : II

Max. Marks : 80

Unit I

Laplace transform of the derivation of $f(t)$, Laplace transforms of integrals multiplication by t , Multiplication by t^n , Division by t use of Laplace transform to unit step function (Heaviside's unit functions) use of Laplace transform to Bessel function, Inverse Laplace transform of derivatives, Convolution, Heaviside's expansion theorem, Problem depends on convolution.

Unit II

Application of Laplace transform to solutions of ordinary Differential equations with constant coefficients.

Unit III

Solution of simultaneous ordinary Differential equations by Laplace transform, Solution of ordinary Differential equations with variable coefficients by Laplace transform.

Unit IV

Solutions of Partial differential Equations by Laplace transform, Application of Laplace transform to integral equation.

Unit V

Heat conduction equations. Problems based on Heat conduction equation using Laplace transform.

Books recommended:-

1. Integral Transform by Goyal & Gupta

Class : M.Sc.
Semester : IV
Subject : Mathematics
Paper Title : Advanced Graph Theory
Paper : III

Max. Marks : 80

UNIT 1

Graphs: Graphs as Models, Paths and connectedness, Cutnodes and Blocks, Graph Classes and Graph Operations, Polynomial Algorithms and NP-Completeness

UNIT II

The Center and Eccentricity, Self Centered Graphs, The Median, Central Paths Path Algorithms and Spanning Trees, Centers.

UNIT III

External Distance Problems: Radius, Small Diameter, Diameter, Long Paths and Long Cycles.

UNIT IV

Convexity: Closure in variants, Metrics on Graphs, Geodetic Graphs, Distance Hereditary Graphs. Diagrams: Diagrams and Connectedness, Acyclic diagrams.

UNIT V

Distance Sequences: The eccentric sequences, Distance sequence, The Distance distribution, Long Paths in Diagrams, Tournaments.

References:

- (1) Bondy and Murthy, Graph Theory with Applications, The Macmillan Press Limited, 1976
- (2) Chartrand G and L. Lesniak, Graphs and Diagrams, Prindle, Weber and Schmidt, Boston, 1986
- (3) Garey M.R, D.S Johnson , Computers and Intractability, A Guide to the Theory of NP-Completeness, Freeman, San Francisco 1979.
- (4) Harary. F, Graph Theory, Addison Wesley Reading Mass 1969 (Indian Edition, Narosa)
- (5) K.R Parthasarathy, Basic Graph Theory, Tata Mc Graw-Hill, Publishing Co, New Delhi, 1994.

Class : M.Sc.
Semester : IV
Subject : Mathematics
Paper Title : Operations Research
Paper : IV

Max. Marks : 80

Unit-I Operations Research - Introduction, Origin and development of Operations Research, Nature and features of Operations Research, Models in Operations Research, General solution methods for Operations Research, Phases of Operation Research, Uses and limitations of operations, Linear Programming Problems : Introduction to Mathematical formulation, Graphical Solution method.

Unit-II General Linear Programming Problem, Theory of Simplex Method, Computational procedure, Numerical problems, Solutions of simultaneous linear equations, Inverse of a matrix using Simplex method.

Unit-III Use of artificial variables, Big-M method, Two phase method, Problem of degeneracy and resolution of degeneracy, Applications of simplex method.

Unit-IV Concept of duality : Introduction, General Primal-Dual pair, formulating a dual problem , Primal-dual pair in matrix form, Economic interpretation of duality, Duality and simplex method, Fundamental Properties and Theorems of duality, complementary slackness, dual simplex method.

Unit-V Post optimality analysis, integer programming, revised simplex method.

Recommended Books:

- 1- Kanti Swarup, P.K. Gupta and Manmohan, Operations Research, Sultan Chand & Sons, New Delhi, 15th revised edition.
- 2- H.A. Taha, Operations Research - An introduction, Macmillan Publishing co. Inc. New York.

Reference Books

1. S.D, Sharma, Operation Research
2. F.S, Hiller and G. J. Lieberman, Industrial Engineering Series, 1995.
3. G. Hadley, Linear programming, Narosa Publishing House 1997.
4. G. Hadley, Linear and dynamic programming, Addison- Wesley Reading mass.
5. H.A. Taha, Operations Research – An Introduction Macmillan Publishing.
6. Prem Kumar Gupta and D.S. Hira, Operations Research, an Introduction S. Chand & Company Ltd. New Delhi.

Class : M.Sc.
Semester : IV
Subject : Mathematics
Paper Title : Theory of Linear Operators
Paper : MMT-405[I]

Max. Marks : 80

Unit-I

Spectral theory in normed linear spaces, resolvent set and spectrum, Spectral properties of bounded linear operators. Properties of resolvent and spectrum. Spectral mapping theorem for polynomials.

Unit-II

Spectral radius of a bounded linear operator on a complex Banach space. Elementary theory of Banach algebras. General properties of compact linear operators.

Unit-III

Spectral properties of compact linear operators on normed spaces. Behaviours of Compact linear operators with respect to solvability of operators equation.

Unit-IV

Fredholm type theorems. Fredholm alternative theorem. Fredholm alternative for integral equation. Spectral properties of bounded self adjoint linear operator on complete Hilbert space.

Unit-V

Positive operators Monotone sequence theorem for bounded self - adjoint operators on a complex Hilbert space. Square roots of a positive operator. Projection operators.

Text Books:

1. E. Kreyszig Introductory functional analysis with applications, John Wiley & Sons, New York, 1978.

Reference Books:

1. P.R. Halmos Introduction to Hilbert space and the theory of Spectral Multiplicity, Second edition, Chelsea publishing co. NY. 1957.
2. N. Dunford and J.T. Schwartz, linear operator -3 part, Interscience/ Wiley, New York 1958-71.
3. G. Bachman and L. Narci, Functional analysis, Academic press New York, 1966.

Class : M.Sc.
Semester : IV
Subject : Mathematics
Paper Title : Fundamentals of Computer Science (Theory and Practical)
Optional Gr. : MMT-405[II]

Max. Marks : 80

Unit-I

Introduction to Object Oriented Programming in C++, Need of Object Oriented Programming, Characteristics of Object Oriented Languages, Class Object and Scopes, Nested classes.

Unit-II

Pointer class member, Class initialization, Constructor and destructor, Assignment and distribution, Virtual functions and Friend functions.

Unit-III

Overloading : Function and operators, Templates, Types of templates and class templates.

Unit-IV

Class Inheritance and their Types, Multiple and Virtual Inheritance.

Unit-V

Operating System: Introduction, What Operating Systems Do; Computer-System Organization, Computer-System Architecture , Operating-System architecture, Operating-System Operations, Process Management, Memory Management, Storage Management, Protection and security, Distributed Systems, Special-Purpose Systems, Client-server Computing , Peer to Peer Computing, Open-Source Operating Systems . (Only Basic Concept of these all.)

Text Books :

1. Robert Lafore Object Oriented Programming in C++, Forth edition, Sams Publishing Indianapolis, IN 46290 USA.
2. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne , Operating System Concept Wiley India Pvt. Ltd, Eights edition.

References:

1. S.B. LIPMAN, , J. LAJOI, C++ PRIMER ADDITION.
2. B.Stroustrup, The C++ programming languages, Addison -Westey.
3. Andrew S. Tanenbaum, Modern Operating System, Pearson International, Third edition.

Class	:	M.Sc.
Semester	:	IV
Subject	:	Mathematics
Paper Title	:	Advanced Special Function-II
Paper	:	MMT-405[III]

Max. Marks : 80

Unit-I Bessel function, Bessel's differential equation, Generating function, Bessel's integral with index half and an odd integer.

Unit-II Generating function for legendre polynomials, Rodrigue's formula, Bateman's generating function, Additional generating functions.

Unit-III Definition of Hermite polynomials $H_n(x)$, Pure recurrence relations, Differential recurrence relations, Rodrigue's formula. Other generating functions, Orthogonality, Expansion of polynomials, more generating functions.

Unit-IV Laguerre Polynomials : The Laguerre Polynomials $L_n(X)$, Generating functions, Pure recurrence relations, Differential recurrence relation, Rodrigue's formula.

Unit-V Jacobi polynomial. Generating functions, Differential Equation of Jacobi Polynomial, Orthogonal Properties.

Books Recommended :-

1. Rainville, E.D., Special Functions, The Macmillan Co., New York 1971.
2. Srivastava, H.M. Gupta K.C. and Goyal, S.P. : The H-functions of One and Two variables with applications, South Asian Publication, New Delhi.
3. Saran, N., Sharma S.D. and Trivedi, - Special Functions with application, Pragati Prakashan 1986.

Reference Books :

1. Lebedev, N.B., Special Functions and Their Applications, Prentice Hall, Englewood Cliffs, New Jersey, USA 1997.
2. Whittaker, E.T. and Watson, G.N., A course of Modern Analysis Cambridge University Press, London, 1963.

Class : M.Sc.
Semester : IV
Subject : Mathematics
Paper title : Fuzzy Sets and their Applications
Paper : MMT-405[IV]

Max. Marks : 80

Unit-I Fuzzy relation on fuzzy sets, Projection and Cylindrical Extensions, Composition of Fuzzy Relation, Min Max Composition and its properties.

Unit-II Binary fuzzy relation, binary relation on a single set, Fuzzy equivalence relation, Fuzzy compatibility relation, Fuzzy relational equation, similarly relation. Fuzzy ordering relation, Fuzzy morphisms.

Unit-III Possibility Theory- Fuzzy measures, Evidence theory, Probability assignment (BPA), Dempster's rule of combined Evidence, Probability measure, Possibility and necessity measure.

Unit-IV Possibility distribution, Possibility theory, Fuzzy sets and Possibility Theory, Versus Probability Theory.

Unit-V Fuzzy logic, Multivalued logic, Fuzzy propositions, Fuzzy quantifiers, Linguistic hedges, Inference from conditional Fuzzy propositions, Inference from conditional and qualified Fuzzy propositions.

RECOMMENDED BOOKS :-

- 1 Fuzzy sets and Fuzzy logic - G.J. Klir and B. Yuan Prentice Hall of India, New Delhi, 1997.
- 2 Fuzzy sets and their applications- Pundir and Pundir, Pragati Prakashan.

REFERENCE BOOKS:-

1. Fuzzy Sets Theory and its Application- H.J. Zimmermann, Allied Publishers Ltd; New Delhi, 1991.
2. Fuzzy sets and Uncertainty and Information- G.J. Kalia Tina A. Folger Prentice Hall of India.

Class : M.Sc.
Semester : IV
Subject : Mathematics
Paper Title : Analytic Number Theory
Paper : MMT-405[V]

Max. Marks : 60

Unit-I

Characters of finite abelian groups, The character group, The orthogonality relations for characters.

Unit-II

Dirichlet characters, Sums involving Dirichlet characters, Dirichlet's theorem on primes in arithmetic progressions.

Unit-III

The half-plane of absolute convergence of a Dirichlet series, the function defined by Dirichlet series, Euler products, The half-plane of convergence of Dirichlet series.

Unit-IV

Analytic properties of Dirichlet series, Mean value formulas for Dirichlet series, Integral formula for the coefficient of Dirichlet series, Integral formula for the partial sums of a Dirichlet series.

Unit-V

Properties of the gamma function, Integral representation of Hurwitz zeta functions, Analytic continuation of Hurwitz zeta functions, Hurwitz formula.

Books Recommended:

1. T.M. Apostol, Introduction to Analytic Number Theory, Narosa Pub. House, 1989.

Class : M.Sc
Semester : IV
Subject : Mathematics
Paper Title : Partial Differential Equations
Paper : MMT-405[VI]

Max. Marks : 80

Unit-I

Formulation, Linear and quasi-linear first order partial differential equations, Paffian equation, Condition for integrability, Lagrange's method for linear equations. First order non-linear equations, method of Charpit - method of characteristics.

Unit-II

Equations of higher order : Method of solution for the case of constant coefficients, Equations of second order reduction to canonical forms, Characteristic curves and the Cauchy problem, Riemann's method for the solution of linear hyperbolic equations, Monge's method for the solution of non-linear second order equations, Method of solution by separation of variables.

Unit-III

Laplace's equations: Elementary solutions, Boundary value problems, Green's functions for Laplace's equation, Solution using orthogonal functions.

Unit-IV

Wave equations: One dimensional equation and its solution in trigonometric series, Riemann-Volterra solution, vibrating membrane.

Unit-V

Diffusion equations: Elementary solution, Solution in terms of orthogonal functions.

Text Books:

1. I.N. Sneddon: Elements of Partial Differential Equations, McGraw Hill, New York, 1957.
2. Ioannis P Stavroulakis, Stepan A Tersian : Partial Differential Equations: An Introduction with MATHEMATICA and MAPLE, World scientific, Singapore, 2004.

Reference Books:

1. P. Prasad, Renuka Ravindran: Partial Differential Equations, New Age International, 1985.
2. T. Amaranath: An elementary course in Partial differential eqations, Narosa Pub., New Delhi, 2009.

Class : M.Sc.
 Semester : IV
 Subject : Mathematics
 Paper Title : Integration Theory
 Paper : MMT-405[VII]

Max. Marks : 80

Unit-I

Definition of a function $f: X \rightarrow Y$ and its properties. One-to-one, onto, bijective functions. Inverse function.

Unit-II

Definition of a set and its properties. Power set, Cartesian product, relations and functions.

Unit-III

Integration of measurable functions.

Unit-IV

Signed measures, Hahn decomposition theorem, Mutually singular measures, Jordan decomposition theorem.

Unit-V

Radon Nykodym theorem, Lebesgue decomposition theorem.

Recommended Books:

1.H.L.Royden, Real Analysis, Mac · lan. p blishing co.Inc,New York,4th edition,1993.

Reference Books:

1. P.R.Halmos, Measure Theory, Van Nostrand.
2. I.K.Rana, Introduction to Measure and Integration, Narosa Publishing House, New Delhi.